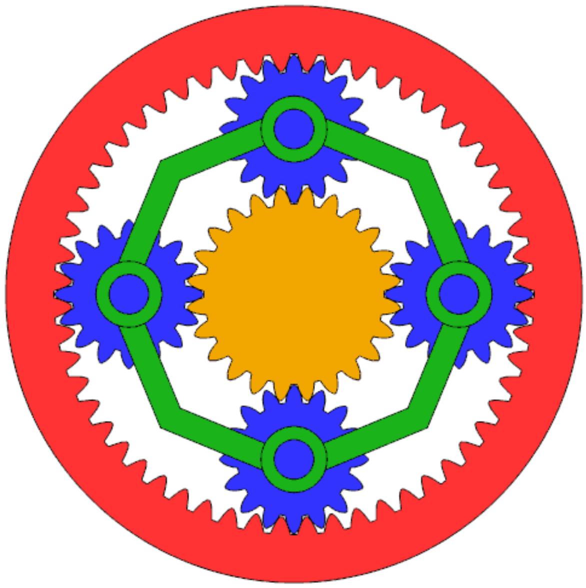

Epicyclic Gear System App

README Document / User Guide

Numerical modelling and simulation tool for load-sharing behaviour of planetary gears

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1 Description

The **Epicyclic Gear System App** is a numerical tool for estimating load-sharing behaviour in single-stage planetary gear systems. It uses user-defined gear geometry, operating conditions, manufacturing errors, input torque, and planet configuration data to calculate system load distribution and related floating-sun response quantities.

The app can be accessed directly at the following link:

https://epicyclic-gear-app-insa-ibmmae-kurtti.streamlit.app/

Key features of the app include:

- load-sharing factor and global load-sharing index estimates;
- floating-sun displacement and tilt outputs;
- optional modelling of selected error, stiffness, eccentricity, and thermal effects;
- sensitivity and Monte Carlo analysis options;
- interactive plots and downloadable numerical results.

2 Purpose and Scope of the App

This app is intended for students, researchers, and engineers studying load-sharing behaviour in real or theoretical planetary / epicyclic gear systems. It provides a practical exploratory interface for comparing how modelling choices and parameter changes affect predicted load distribution.

2.1 Project Background

The project was initially developed in response to a CMDGears request for an analytical tool capable of estimating simplified load-sharing factors in planetary gear units with an arbitrary number of planets. The original scope focused on geometrical defects, mechanical stiffness, and sun gear orbital motion. The current version extends this work into a more complete web-based app interface while retaining the reduced-order engineering approach.

2.2 Model Scope, Assumptions, and Limitations

The solver uses a quasi-static reduced-order formulation. It is designed for analytical comparison of load-sharing trends, rather than for gearbox certification, experimental validation, or detailed local stress prediction.

The main assumptions and limitations include:

- the system is treated as a single-stage planetary gear system;
- dynamic effects, vibration, and time-dependent inertial effects are not explicitly modelled;

- finite-element tooth-contact stresses, tooth-root fatigue, and contact fatigue are not calculated;
- thermal effects are represented using a reduced engineering approximation;
- results should be interpreted as analytical estimates based on the selected inputs and modelling assumptions.

For the full theoretical background, mathematical formulation, and validation discussion, please refer to the accompanying final project report, *Numerical Modelling Tool for Load Sharing Behavior of Epicyclic Gears*, available as a downloadable `PDF` from the app.

3 App Structure

The app is structured into three main parts:

1. **User Inputs** – the section where the model parameters are defined.
2. **Numerical Results** – the section containing the main numerical outputs of the solver.
3. **Plots** – the visual output section, organized into several categories. Depending on which options are enabled in the input panel, some plots may appear or remain hidden.

The main plot categories are:

1. **System Overview**
2. **Error Decomposition**
3. **Phase Response & Load Sharing**
4. **Planet-Specific Forces**
5. **Monte Carlo Analysis**
6. **Structural Trends**
7. **Sensitivity Analysis**

4 General Workflow

A typical run of the app follows the workflow below.

4.1 Step 1: Set the Basic Inputs

Start by entering the main system parameters in the user input section, including the number of planets, input torque, sun radius, pressure angle, and bearing type.

4.2 Step 2: Enable Optional Geometry Settings

If the default geometry settings are sufficient, the geometry options can be left unchanged.

If a custom gear geometry is required, enable the corresponding geometry option and enter the desired tooth numbers, module, helix angle, centre distance, face width, profile shifts, or pressure-angle settings.

4.3 Step 3: Enable Advanced Effects if Needed

Advanced effects are disabled by default. They should only be enabled when the user specifically wants those effects to influence the calculation.

Advanced options may include:

- **Error and geometry-related effects**
  - manufacturing or assembly error presets;
  - user-defined planet errors;
  - periodic eccentricity.
- **Stiffness and operating-condition effects**
  - periodic mesh stiffness;
  - support stiffness options;
  - reduced thermal effects.
- **Study and analysis options**
  - sensitivity analysis;
  - Monte Carlo analysis.

4.4 Step 4: Run the Analysis

After selecting the desired inputs, press the ▶ **Run Analysis** button.

The app will execute the solver using the current inputs and generate the corresponding numerical outputs and plots.

4.5 Step 5: Inspect the Numerical Results

The numerical results section summarizes the main outputs of the run. These include the final load-sharing values, global load-sharing index, floating-sun displacement and tilt, and other relevant solver outputs.

4.6 Step 6: Inspect the Plots

After reviewing the numerical results, use the plot categories to examine planet-specific behaviour, phase-dependent response, error contributions, and optional sensitivity or Monte Carlo trends.

4.7 Step 7: Download Results if Needed

The app includes an option to download numerical results as a CSV file. This can be useful for further analysis, post-processing, report writing, or comparing multiple simulation cases.

5 Output Definitions

This section gives brief descriptions of the main outputs shown in the app.

5.1 Planet Load-Sharing Factor

The planet load-sharing factor, or LSF, indicates how much load a given planet carries relative to the ideal equal-load condition. An LSF value close to 1 indicates that the planet is carrying approximately its ideal share of the load. Values above 1 indicate that the planet is carrying more than its ideal share, while values below 1 indicate that it is carrying less.

5.2 Global Load-Sharing Index

The global load-sharing index, usually denoted as K_γ, describes the maximum load-sharing imbalance in the system. It is based on the most highly loaded planet relative to the ideal average planet load. A higher value of K_γ indicates a larger load concentration on one or more planets.

5.3 Force Distribution

The force distribution outputs show the load carried by each planet gear, allowing direct comparison between planet branches.

5.4 Worst Phase

The worst phase corresponds to the rotational phase position at which the system reaches its maximum load-sharing imbalance. Several final output plots are shown at this worst-phase condition.

5.5 Floating-Sun Displacement and Tilt

The floating-sun outputs describe the calculated displacement and tilt response of the sun gear in the reduced-order model. These outputs help indicate how the sun gear moves or tilts in response to unequal planet loading, stiffness variation, and geometric mismatch.

5.6 Error Components

The error decomposition plots show how different error sources contribute to the equivalent error at each planet. These may include pin-position effects, runout, rigid offsets, profile errors, periodic eccentricity, or thermal contributions, depending on the selected inputs.

5.7 Sensitivity Analysis

The sensitivity analysis option performs parameter sweeps to show how selected model outputs change when certain parameters are varied. This is useful for identifying which inputs have the strongest influence on the load-sharing behaviour.

5.8 Monte Carlo Analysis

The Monte Carlo option repeats the analysis over multiple error realizations. This allows the user to inspect statistical trends and variability in the predicted system response.

6 Special App Features

Several interface features are included to make the app easier to use.

6.1 Input Controls

6.1.1 Conditional Input Fields

Several advanced input fields are greyed out by default. These fields become editable only after the corresponding checkbox has been selected. This is intentional; it ensures that optional effects are only included when the user explicitly enables them. If an optional feature is not selected, the solver uses the default values defined for standard runs.

6.1.2 Collapsible Input Section

Additionally, the entire user input section is collapsible. This allows the user to hide the input panel after running an analysis and focus on the numerical results and plots.

6.2 Interactive Plot Controls

The plots available in the Plots section are fully interactive. Users can zoom, pan, hover over data points, and inspect values directly.

6.2.1 Legend Behavior

Legend items can be used to control the displayed traces:

- clicking a legend item hides or shows that trace;
- double-clicking a legend item isolates that trace;
- clicking directly on plotted elements can visually de-emphasize or reselect them, depending on the plot interaction mode.

6.2.2 Opening and Downloading Plots

Plots can be opened separately and downloaded as image files (`PNG`). This can be useful for preparing reports, presentations, or documentation.

6.2.3 Conditional Plot Display

Certain plots are only displayed when the required input options are enabled. This keeps the interface cleaner and prevents unused plots from cluttering the app. If a plot is not available, the app provides a message explaining which setting or parameter must be activated to generate it. For example, Monte Carlo plots require the Monte Carlo checkbox to be selected in the user inputs.

6.2.4 Dropdown Plot Options

Some plot sections include dropdown menus. These allow the user to inspect different output quantities, response formats, or visualization modes within the same plot category.

6.3 Running Multiple Analyses

The app does not need to be reloaded between runs. To run a new case, edit the desired parameters and press ▶ **Run Analysis** again.

6.4 Device Compatibility

The app can be opened and operated from any device with a web browser. However, for detailed plot inspection and larger parameter studies, a laptop or desktop screen is recommended.

7 Project Files and Repository

The online version of the app can be used without local installation, GitHub access, or direct access to the source code. The full GitHub repository is provided separately for users who want to inspect, modify, or run the implementation locally.

The current version retains the solver logic and mathematical framework of the previous MATLAB implementation, while adapting the project for a web-based Streamlit interface with improved accessibility and usability. The backend solver is implemented in Python.

The main files in the repository are:

File	**Purpose**
`app.py`	Streamlit frontend code used to build and display the web-based app interface.
`epicyclic_gear_system.py`	Numerical solver backend used to run the calculations.
`requirements.txt`	List of Python packages required for the Streamlit app to run independently.
`Final Report PDF`	Complete technical report containing the mathematical formulation, validation discussion, assumptions, and limitations associated with the model and solver.
`GitHub README`	Short repository-level description of the app and its purpose.

8 Recommended Use

For a standard use case, the user should begin with the default settings or with a simple controlled input case. Optional effects should then be activated one at a time so that their influence on the results can be understood clearly.

For example, a user may first run a baseline case, then add selected error sources or advanced effects depending on the objective of the study. This approach makes it easier to interpret how each modelling choice affects the final load-sharing behaviour.